UNIVERSITY OF KWAZULU-NATAL ™
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PRINCIPLES AND GUIDELINES ON THE USE OF GENERATIVE ARTIFICIAL INTELLIGENCE TOOLS IN ACADEMIC WORK

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1. Purpose

The purpose of the *AI Academic Guidelines* is to establish clear principles and practical guidelines for the use of generative artificial intelligence (AI) tools in academic work at the University of KwaZulu-Natal (UKZN). With the rapid evolution of AI technology and the implications for higher education sector, it is essential to provide clear guidance that ensures AI tools are used ethically, responsibly, and in a manner that upholds academic integrity.

By establishing these principles and guidelines, UKZN aims to create a framework that not only addresses the ethical and practical considerations of generative AI use in academia but also aligns with national and international standards for AI deployment in educational and research settings. This document serves as a dynamic resource that can be updated as AI technology continues to evolve, ensuring that UKZN remains at the forefront of integrating generative AI into academic work in a responsible and impactful way.

2. Introduction and background

The integration of generative artificial intelligence (AI) tools into academic work is transforming the landscape of higher education and research globally. Generative AI refers to artificial intelligence systems that can create new content, such as text, images, audio, or code, based on patterns learned from large datasets. These AI tools, like ChatGPT, can generate human-like responses to prompts, assist with various tasks, and produce creative outputs. Moreover, modern generative AI tools are increasingly conversational and context-aware, engaging with users in human-like dialogues. They are deeply integrated into various digital platforms, often operating seamlessly in the background, which makes distinguishing between AI-assisted and purely human-generated content more challenging. The advent of voice-activated AI and conversational interfaces further blurs the lines between AI use and non-use in academic work.

As AI continues to evolve, it offers unprecedented opportunities to enhance the way knowledge is created, disseminated, and applied. However, the rapid advancement of AI also presents significant challenges, particularly concerning the ethical and responsible use of these technologies, the preservation of academic integrity, and the need for appropriate guidance on their implementation. As generative AI tools become more prevalent, it is crucial to promote not only ethical considerations but also the responsible use of these technologies. Responsible use encompasses practical aspects

of integrating AI into academic work while maintaining accountability, transparency, and adherence to academic standards.

At UKZN, the use of generative AI in academic work is increasingly prevalent among both academic staff and students, with majorities of these groups reporting that they already use generative AI in their academic work. Recognising the profound impact that generative AI can have on teaching, learning, and research, UKZN has embarked on the development of these *AI Academic Guidelines*. This initiative is rooted in the understanding that while generative AI tools can greatly enhance academic productivity and creativity, their use must be governed by clear principles that ensure responsible and ethical practices.

The *AI Academic Guidelines* are designed to provide a comprehensive framework for the use of generative AI tools in academic work, addressing both the opportunities and challenges they present. It draws on the insights gathered from a survey conducted among UKZN's academic staff and students, which highlighted the growing reliance on generative AI tools for various academic purposes and the need for structured guidance on their appropriate use. The survey results indicated a strong demand for guidance that spans general principles and specific applications, reflecting the diverse needs of the academic community.

In developing the *AI Academic Guidelines*, UKZN is also aligning with the South African National AI Policy Framework, which emphasises the importance of using generative AI to drive innovation whilst upholding ethical standards and academic rigour. The *AI Academic Guidelines* seek to ensure that generative AI is used in ways that are consistent with these national priorities.

Moreover, the *AI Academic Guidelines* support UKZN's strategic goals of Excellent Teaching and Learning, Excellent Student Experience, and Excellent and High Impact Research, Innovation, and Entrepreneurship. Furthermore, in developing the *AI Academic Guidelines*, UKZN had the benefit of being able to review examples of similar documents by other universities in South Africa and internationally, to ensure that the quality of the *AI Academic Guidelines* is of a high standard.

The *AI Academic Guidelines* serve not only as a set of rules but as a dynamic resource that will evolve as AI technology advances. It is intended to be a living document, regularly reviewed and updated to reflect new developments in AI and its applications in

academia. By providing clear and practical guidance, UKZN aims to empower its academic community to harness the potential of generative AI whilst maintaining the highest standards of academic integrity and responsibility.

3. Definition of terms

Academic work:	Teaching, learning, research and community engagement.
AI:	Artificial Intelligence is the simulation of human intelligence in machines that are programmed to think, learn, and respond like humans, enabling them to perform tasks that typically require human intelligence and problem-solving abilities.
Exit-level module:	A module at the highest level required by the Higher Education Qualifications Framework (HEQF) for a qualification.
Generative AI:	AI systems that can create content—such as text, images, audio, or code—based on patterns learned from large datasets. These systems often engage conversationally and contextually with users, enabling interactive dialogues that go beyond simple content creation.
Prompt:	The human input to guide the generative AI's output.
Student work:	Theses, dissertations, and assignments.
Secure generative AI tool:	A secure generative AI tool is a system that operates in a fully offline environment, ensuring that all data inputs, outputs, and processing occur on local hardware without external connectivity. This design guarantees that user data is neither shared nor accessible to developers or external entities, preserving privacy and control over confidential information.

4. Scope

The *AI Academic Guidelines* apply to all academic work.

5. Principles

UKZN commits itself to four inter-dependent principles for the use of generative AI tools in academic work:

5.1. Use AI to innovate and advance knowledge

The academic community should harness the power of generative AI to drive innovation and push the boundaries of knowledge. Generative AI should be leveraged to explore new ideas, solve complex problems, and contribute to meaningful advancements in various fields. Generative AI should be a catalyst for creativity and interdisciplinary collaboration, propelling research and scholarship to new heights.

Alignment with the South African National AI Policy Framework:

This principle aligns with the South African National AI Policy Framework's emphasis on using AI to drive innovation and contribute to the country's socio-economic development. By integrating AI into academic research, UKZN supports the national goal of maintaining South Africa's competitive edge in the global knowledge economy.

5.2. Use AI ethically and responsibly

The academic community must embrace generative AI with a commitment to ethical practices, ensuring that generative AI tools are used in ways that uphold fairness, transparency, and accountability. Generative AI tools should also be used in a way that respects privacy. Generative AI should never replace human critical thinking and creativity, which remain central to the integrity and quality of academic work. Responsible use entails being mindful of the implications of generative AI outputs, verifying the accuracy of information provided by generative AI tools, and being accountable for how AI-generated content is integrated into one's work. This includes critically evaluating generative AI contributions, avoiding over-reliance on generative AI at the expense of developing one's own skills, and ensuring that generative AI use aligns with these *AI Academic Guidelines*.

Alignment with the South African National AI Policy Framework:

This principle reflects the South African National AI Policy Framework's focus on ethical generative AI usage, emphasising the need for fairness, transparency, and accountability in all AI applications. Ethical use involves considering how AI affects not just the individual user but also society at large, ensuring that AI applications do not perpetuate biases or harm. It also aligns with the framework's advocacy for responsible generative AI deployment that complements rather than replaces human decision-making.

5.3. Uphold academic rigour and integrity

The academic community should maintain the highest standards of academic integrity by ensuring that substantial generative AI contributions are methodologically sound and appropriately acknowledged. Generative AI should be used in a way that enhances, rather than diminishes, the quality and credibility of scholarly work.

Alignment with the South African National AI Policy Framework:

This principle aligns with the South African National AI Policy Framework's emphasis on the ethical use of AI across various sectors. In the context of academic work, ethical use requires—over and above the considerations mentioned under the previous principle—academic rigour and integrity. It involves proper attribution, avoiding plagiarism, adhering to academic standards, thus ensuring that the use of generative AI does not compromise the credibility and validity of scholarly work.

5.4. Build capacity and foster education

Generative AI literacy and skills should be developed across all levels. Efforts should be made to empower both students and staff to use generative AI tools effectively and responsibly, enhancing academic capabilities throughout the institution. Generative AI literacy and skills must go hand in hand with fostering human critical thinking and creativity to ensure that technology enhances, rather than replaces, the essential role of human judgement in academia.

Alignment with the South African National AI Policy Framework:

The South African National AI Policy Framework prioritises the integration of AI into educational curricula from basic education to tertiary levels. By building generative AI capacity at UKZN and incorporating generative AI into the university's teaching and learning programmes, the institution aligns with the national objective of preparing students for the AI-driven future and fostering a society that is proficient in AI technologies. This integration supports the broader goal of leveraging generative AI for national development and economic growth.

6. Guidelines

6.1. Basic guidelines

6.1.1. Responsible use of generative AI is encouraged

The use of generative AI is encouraged, provided that it aligns with these *AI Academic Guidelines.* In particular, when using generative AI, users always remain responsible for the final content of their academic output. In the context of team-based academic work, each group member is individually responsible for ensuring that their use of generative AI aligns with the *AI Academic Guidelines*.

6.1.2. The limitations of generative AI should be recognised

Generative AI has various limitations that highlight the importance of using it as a supplement to, rather than a substitute for, human intelligence in academic work. These limitations include a lack of access to current research and real-time updates; the potential for "hallucinations" or generating false information; and an inability to conduct its own research or understand complex academic contexts. Generative AI encompasses various models, including language models, image generators, audio synthesisers, and code generators. While many generative AI tools are language-based, it is important to recognise that generative AI spans multiple content types. Users should be aware of the specific capabilities and limitations of the generative AI tool they are using.

Generative AI also struggles with bias in training data and the influence of embedded biases. The responses generated by AI tools are influenced not only by the underlying models and the data they have been trained on but also by human-designed guardrails and biases embedded in the system. This may result in outputs that reflect specific cultural, societal, or ideological perspectives, which may not be universally applicable or accurate. Furthermore, generative AI lacks critical thinking and ethical judgment, and may misinterpret language, cultural nuances, and interdisciplinary integration.

Additionally, generative AI may provide incorrect citations, fail to grasp tacit knowledge, and offer limited explanations for its reasoning. These factors underscore the necessity of academic rigour—and taking full responsibility—when integrating AI into academic environments.

6.1.3. Academic rigour is key

Academic rigour in the use of generative AI necessitates critically evaluating AI outputs, especially considering their limitations. Academic staff and students must verify AI-generated content against credible sources to ensure accuracy. When using ideas suggested by generative AI, it is essential to be familiar with the relevant literature to properly attribute existing concepts. Furthermore, academic rigour demands transparency in methodology and proper attribution, as outlined in these AI Academic Guidelines. When AI is integral to a study's methodology, providing detailed documentation to ensure reproducibility further upholds the scholarly value of work involving AI.

6.2. Academic integrity

6.2.1. Avoiding AI misrepresentation

Misrepresenting the role of generative AI in academic work by failing to disclose the use of generative AI in one's work as provided for in these *AI Academic Guidelines,* or failing to do so in an accurate and truthful matter constitutes academic misconduct. It can mislead educators and peers about the extent of one's actual contribution to the academic work, which undermines academic integrity and is thus seen in a serious light.

6.2.2. Avoiding plagiarism

Generative AI does not possess its own agency or creativity, and any content it generates is a direct result of the prompts given by the user. Therefore, using AI-generated text is not plagiarism in itself. However, users must be vigilant about the origins of the ideas and information produced by generative AI. Generative AI draws from vast datasets, which include previously published content. As a result, the ideas or information it presents may already belong to someone else. Users must be aware of this risk and take responsibility for conducting their own thorough literature searches to ensure that all ideas presented, even those generated by AI, are properly credited to their original sources, where applicable. Failing to do so and using someone else's ideas without proper attribution—even if those ideas are presented by generative AI—constitutes plagiarism and possibly copyright infringement. As stated in paragraph 6.1.3 above, when using ideas suggested by generative AI, it is essential to be familiar with the relevant literature to properly attribute existing concepts.

6.3. Guidelines for key areas of academic activity

6.3.1. Academic reading

Generative AI can assist with summarising articles, outlining key points, and generating questions that prompt deeper understanding and engagement with the text. However, these tools are designed to supplement, not replace, without undermining the development of critical reading skills. Users should actively engage with texts, critically evaluate AI-generated content, and ensure that their academic development remains rooted in independent thought and comprehensive analysis. Reliance on generative AI alone may lead to a superficial understanding.

6.3.2. Academic writing and presentation

Generative AI can be used to, among others, refine language, improve presentation, and generate drafts with suggested structure and clarity. The onus is on the user to critically engage with any AI-generated material to ensure that each statement generated is factually accurate, not plagiarised from other works, and properly referenced.

6.3.3. Data analysis

Generative AI can assist with data analysis by providing preliminary interpretations, identifying patterns, and suggesting potential models or hypotheses. However, users must maintain full control over the analytical process, exercising caution and ensuring that AI-generated insights are rigorously evaluated and validated using statistical methods and cross-verification. While generative AI can be a powerful tool, it has limitations, particularly in handling numerical data and interpreting nuanced insights. Therefore, generative AI should be used to enhance, not replace, the user's expertise and critical thinking. To ensure accuracy, reliability, and integrity of research findings, final analyses should always involve thorough manual validation.

6.3.4. Supporting learning

Generative AI can be used to organise notes, create study aids, and generate practice questions tailored to individual needs. Generative AI can also be used to translate text between languages to ensure better understanding of concepts. Translation abilities may vary between languages. It must be emphasised that the AI-generated study materials must be verified by credible published works, and used to enhance, rather than replace, active learning and comprehension.

6.3.5. Supporting teaching

Generative AI can be utilised to enhance educational content creation. This includes assisting with lesson plans, generating new examples or real-world applications to make lectures more engaging and relevant, providing feedback to lecturers on their lesson plans and lectures, and enhancing tutorial offerings, assessment, and feedback. However, it is crucial to emphasise that generative AI should enhance rather than replace the role of the lecturer. The use of generative AI as a sole tutor or for marking purposes presents significant challenges and should be approached with caution.

6.3.6. Developing coding skills

Generative AI can be a valuable resource for coding, but it should be used as a complementary tool, not as a primary solution. Students must write and test their own code to develop the fundamental coding skills that underpin the structure and logic of the task or output. Whilst generative AI can suggest code, explain complex concepts, and assist with debugging, it is essential to review and test any AI-generated code carefully, as it may be outdated or incorrect. This approach ensures that students build strong coding abilities while responsibly leveraging AI tools.

6.4. Integration of generative AI skills into the curriculum

6.4.1. Alignment with national and institutional priorities

UKZN is committed to integrating generative AI literacy and skills into its curriculum across all schools and programmes. This integration aligns with the South African National AI Policy Framework's emphasis on embedding AI education into the broader educational landscape. It also supports UKZN's dedication to preparing students for the future of work and research within the African context.

6.4.2. Curriculum content areas

The curriculum should address key generative AI skill areas, including but not limited to:

- Generative AI fundamentals and terminology
- Data literacy and management
- Machine learning basics
- Generative AI ethics and responsible use

- Limitations, potential biases and challenges of generative AI use
- Applications of generative AI in specific disciplines
- Critical evaluation of generative AI outputs
- The impact of generative AI on society and the workplace

6.4.3. Approaches to curriculum integration

Schools are encouraged to adopt one or both of the following approaches to integrating AI content in their curricula:

1. **Self-standing generative AI literacy and skills modules:** Develop one or more new, self-standing modules that provide generative AI skills, as per paragraph 6.4.2 above. Schools within a College may consider developing common modules, where appropriate.
2. **Integration into existing modules:** Incorporate generative AI skills, as per paragraph 6.4.2 above, into current modules across various disciplines.

6.4.4. Emphasis on complementary skills

When integrating generative AI skills into the curriculum, Schools should emphasise how these skills complement traditional academic skills such as critical thinking, problem-solving, and data analysis. Generative AI learning should be contextualised within African scholarship and local challenges, with a strong focus on ethical considerations and responsible generative AI use. Hands-on experience with generative AI tools relevant to each discipline should be provided, and interdisciplinary collaboration in AI-related projects and research should be encouraged.

6.4.5. Ongoing training and curriculum review

UKZN should provide continuous training and support for all staff involved in student training to effectively integrate generative AI skills into their teaching. The generative AI curriculum should be reviewed and updated annually to keep pace with the rapid advancements in AI technology and its applications.

6.5. Module-specific rules on the use of generative AI tools

6.5.1. Flexibility in AI tool usage

Module coordinators have the discretion to implement specific rules regarding the use of generative AI tools within their modules. This flexibility allows for the customisation of generative AI tool usage based on the unique requirements and

learning objectives of each module. Whilst UKZN encourages the use of generative AI in academic work as a general guideline, there can be good pedagogical reasons to limit the use of generative AI in certain modules. In such cases, module coordinators and their discipline Academic Leaders are empowered to limit the use of generative AI in their modules, or even ban it altogether.

6.5.2. Levels of AI tool usage

Module coordinators may specify different levels of generative AI tool usage within their modules, including:

- **Full use:** Generative AI tools can be used for all aspects of the module, provided there is proper disclosure, as provided for in paragraph 6.8 below. Note that if no rules are specified for a module, this will automatically be the default position, aligned with the general guideline that the use of generative AI is encouraged in academic work.
- **Limited use:** Generative AI tools can be used for specific tasks, provided there is proper disclosure, as provided for in paragraph 6.8 below, but are restricted for others.
- **No use:** Generative AI tools are not permitted for any aspect of the module.

6.5.3. Developing rules

When developing rules for the use of generative AI tools within their modules, module coordinators should:

- Clearly define the permitted and prohibited uses of generative AI tools in the module outline.
- If limitations are placed on the use of generative AI, or if not permitted at all, provide the pedagogical rationale for such limitation or ban in the module outline.
- Ensure that the rules align with the module's learning outcomes and assessment criteria.
- Consider the potential impact on students' skill development and academic integrity.

6.5.4. Approval and communication process

The process for developing module-specific rules on generative AI tool usage should include:

- **Development:** Rules should be created by the module coordinator.

- **Review and approval:** The Academic Leader of the discipline should review and approve the rules.
- **Communication:** The approved rules must be clearly communicated to students in the module outline, discussed at the beginning of the module, and accompanied by examples of appropriate and inappropriate uses of generative AI specific to the module.

6.5.5. Ensuring consistency and fairness

While module-specific rules are encouraged, discipline Academic Leaders must ensure consistency in generative AI usage rules across related modules within a programme, particularly where modules share comparable learning objectives and assessments. This approach helps maintain coherence and fairness in AI education within the broader curriculum. In cases where a module prohibits the use of generative AI, the discipline Academic Leader must ensure that students still receive generative AI literacy and skills training through other modules or dedicated workshops within the programme. This ensures that all students gain essential AI competencies, regardless of individual module restrictions.

6.5.6. Annual review and updates

Module-specific rules on the use of generative AI tools should be reviewed annually to ensure they remain relevant and effective in light of technological advancements and evolving educational needs. Any limitations on generative AI tool usage must be well-explained in the module outline and should never be arbitrary.

6.6. Using generative AI in student assessment

6.6.1. Role of generative AI in examination setting

Module coordinators and examiners may use generative AI tools to assist in generating examination questions, model answers, and structuring examination papers. However, generative AI should serve to enhance—not replace—the expertise and judgement of academic staff in the exam-setting process.

6.6.2. Ensuring originality and relevance

When using generative AI to assist in setting exam papers, examiners should:

- Use AI-generated content as a starting point or source of inspiration, ensuring that final exam questions are modified and contextualised to align with the specific module content and learning objectives.

- Critically evaluate generative AI-generated content to ensure it aligns with the module's learning outcomes and assessment criteria.
- Modify AI-generated questions to reflect the appropriate depth and breadth of knowledge expected at the relevant academic level.

6.6.3. Preserving academic rigour

Examiners should:

- Review AI-generated questions for accuracy, clarity, and appropriate difficulty level.
- Ensure that questions assess higher-order thinking skills rather than just factual recall.
- Adapt AI-generated content to reflect local contexts, African perspectives, and UKZN's academic standards.

6.6.4. Documentation

Examiners who use generative AI in setting examination papers must:

- Document the use of generative AI tools during the examination paper preparation process.
- Specify which sections or questions were initially generated by generative AI and describe how they were modified and reviewed as per these *AI Academic Guidelines.*
- Retain records of the generative AI prompts used and the original AI outputs.
- Make these documentation available to the module coordinator and discipline academic leader, and upon request, also to any moderator.

6.6.5. Quality assurance

Examination papers set using generative AI must undergo the same rigorous review and approval process as traditionally set papers. Examiners who use generative AI in setting examination papers must notify the relevant module coordinator. The module coordinator must, in turn, notify the moderators, where applicable. The module coordinator and the moderators, where applicable, may request to peruse the documentation contemplated in paragraph 6.6.4 above. Upon such request, the examiner must provide such documentation. The purpose is to allow the module coordinator and the moderators, if any, to ensure that these *AI Academic Guidelines* are adhered to. For exit-level modules or where external examination is required:

- The module coordinator must also notify the external examiner of the use of generative AI in setting the examination paper.

- The external examiner may also request to review the documentation related to the use of generative AI, as outlined in paragraph 6.6.4.
- The examiner must provide this documentation to the external examiner upon request, to ensure transparency and adherence to these *AI Academic Guidelines*.

6.6.6. AI-assisted marking

Module coordinators may decide, subject to approval by the discipline Academic Leader, to use generative AI tools to assist in marking assessments in a particular module. This may range from providing initial feedback to fully automated marking. However, the ultimate responsibility for the accuracy and fairness of marking always remains with the examiner. When using generative AI for marking, examiners should:

- Ensure the AI tool is calibrated to specific assessment criteria and rubrics.
- Critically review all AI-generated feedback or marks before finalisation.
- Maintain oversight to catch potential errors or biases.
- Use AI-generated feedback as a basis for more detailed, personalised feedback.
- Regularly evaluate the effectiveness and fairness of AI-assisted marking.
- Be transparent with students about the use of generative AI in marking.
- Ensure that generative AI use does not compromise the detection of academic misconduct.

6.6.7. Approval and review of AI-assisted marking

The implementation of AI-assisted marking should be approved by the relevant discipline Academic Leader, align with UKZN's assessment regulations, and undergo regular review to ensure the integrity and quality of marking are maintained.

6.6.8. Restrictions on AI-assisted marking

The use of generative AI for marking final exit-level examinations is not permitted without approval from the University Senate. This guideline will be subject to regular review as AI technology and educational practices evolve.

6.7. Using generative AI in student supervision

6.7.1. General

The integration of generative AI tools into student supervision offers new possibilities for enhancing the quality and efficiency of support provided to students. At UKZN, supervisors are encouraged to consider the following guidelines when incorporating generative AI tools into their supervision practices:

6.7.2. Academic integrity

Generative AI should serve as a supplement to the supervisor's expertise rather than a replacement. While generative AI can assist with tasks like drafting feedback, suggesting resources, or summarising complex information, supervisors should ensure that they retain final responsibility for the quality and integrity of supervision.

6.7.3. Disclosure

Supervisors are encouraged to disclose, as a once-off in their supervision agreements, how they intend to use generative AI in their supervision. Should this approach change, supervisors should inform students accordingly in writing. This includes disclosing to students how generative AI tools will be used, providing transparency about the role generative of AI in feedback and decision-making processes.

6.7.4. Fostering AI literacy

Supervisors should foster AI literacy among students, encouraging them to develop skills for using AI tools responsibly and effectively. This includes guiding students on verifying AI-generated content, recognising potential biases, and incorporating AI-generated insights into their work with academic rigour.

6.8. Disclosing the use of generative AI in research and research-related activities

6.8.1. Application

These guidelines on disclosing generative AI use apply to formal academic work by academic and students related to: (a) research, and (b) research-related activities, such as peer-review. This encompasses academic outputs such as articles in academic journals, academic books, and peer-review reports. Disclosure of generative AI use is not required in informal academic work related to research,

where the focus is less on advancing knowledge in the academic discipline, but rather on communicating with a broader audience, such as editorials, blog posts, podcasts, public reports, and video abstracts.

The use of generative AI in student work, is addressed separately under paragraph 6.9 below.

6.8.2. These guidelines as a basic standard

These guidelines establish a basic standard for the ethical use of generative AI in research. However, if a user submits an article to an academic journal or writes a book, blog post, or other content where the publisher requires more detailed disclosure of generative AI use than these guidelines, they must adhere to the publisher's more stringent requirements.

6.8.3. Levels of disclosure

For articles in academic journals and academic books, three levels of disclosure are distinguished:

Level 1: Optional disclosure

If generative AI tools are used in a way that does not significantly contribute to the intellectual content of the research—such as improving language or compiling a reference list—disclosure is not necessary. These uses of generative AI are comparable to the work of human language editors and research assistants, where there is no ethical obligation to acknowledge their contributions. For consistency, the same principle should apply to generative AI tools used for these purposes.

Level 2: Disclosure of use, not prompts

When a generative AI tool is used in a way that significantly contributes to the intellectual content of the research but does not impact the methodology—such as generating ideas or initial drafts—its use should be disclosed. However, the prompts and materials provided to the generative AI do not need to be disclosed.

Level 3: Disclosure of use and prompts

When generative AI tools are used in a way that not only significantly contributes to the intellectual content of the research but also directly impacts the validity and reliability of the research—meaning they are relevant to the methodology—both the

use of generative AI and the prompts/materials provided to the AI should be disclosed. Examples of generative AI use relevant to the methodology include:

- AI analyses and interprets quantitative data, shaping research conclusions.
- AI interprets qualitative data, impacting research findings.
- AI develops predictive models that directly influence research outcomes.
- AI contributes to forming hypotheses central to the research design.
- AI aggregates and analyses data from multiple sources, shaping research conclusions.
- AI simulates scenarios that directly influence research outcomes.

6.8.4. Details to be disclosed

When disclosing the use of a generative AI tool, it is best practice to provide the following details: the name of the AI tool, its version, its creator (if it is a lesser-known AI tool), its URL (if accessed online and not as a desktop application), and the purpose of its use (e.g., to generate an initial draft). There is no need to disclose the specific dates of use.

6.8.5. Where to disclose

When generative AI use relevant to the methodology is disclosed, it should be included in the methodology section. This ensures that the role of generative AI in shaping the research's validity and reliability is clear. The specific prompts and materials provided to the generative AI can be detailed within this section or, if more appropriate, included as supplementary files. If generative AI use that is not relevant to the methodology is disclosed, an acknowledgment section should be included in which the use of AI is disclosed. This keeps the disclosure visible whilst ensuring that the methodology section remains focused on methodological details.

6.8.6. AI tools cannot be listed as co-authors

Generative AI tools, such as ChatGPT-4, are not considered authors and should not be credited as co-authors on any research output. They lack the capacity for intellectual responsibility and accountability, which are essential qualities for authorship.

6.8.7. AI tools cannot be referenced as sources

If generative AI provides information or ideas that are incorporated into your work, it is your responsibility to verify and locate the original sources of that information or ideas. AI tools should not be cited as sources; instead, you should cite the original

sources that the AI might have drawn from. If the original source cannot be identified, exercise caution and consider whether to include that information in your work.

6.9. Disclosing the use of generative AI in student work

6.9.1. Full disclosure

Any use of generative AI tools in the creation of student work, from generating ideas to editing or refining text, must be explicitly disclosed. This is crucial for assessing the student's individual contribution and for understanding the role of generative AI in shaping the final product.

6.9.2. Record-keeping as a learning tool

Supervisors (for theses and dissertations) and module coordinators (for assignments) may require students to keep records of their substantial interactions with generative AI tools, such as the prompts provided and the AI-generated outputs or responses. This record-keeping serves as a learning tool, helping students reflect on their use of generative AI and understand its impact on their work. Keeping such records and discussing them with educators provides an opportunity to deepen the student's learning experience. It allows for a critical evaluation of how generative AI tools influence thought processes, decision-making, and the development of ideas.

6.9.3. Promoting reflective learning

Requiring students to maintain and review AI prompts and outputs is not a punitive measure but an educational opportunity. This practice encourages reflective learning, enabling students to critically assess their reliance on generative AI and develop a more nuanced understanding of its capabilities and limitations. Discussions around these records can help students improve their research and writing skills, ensuring that they are not only proficient in using generative AI tools but also capable of producing independent and original work.

6.9.4. Where to disclose AI Use

All student work must include the standard generative AI disclosure form in **Appendix A** in the front section of the assignment, dissertation, or thesis. This ensures consistency in how AI use is reported by students and allows for a transparent assessment of the student's work.

6.10. When to use secure generative AI tools

Generally, most academic tasks do not necessitate secure generative AI tools. However, users are encouraged to configure these tools to avoid using inputs for training purposes, enhancing data privacy. When handling sensitive personal information or protectable intellectual property—especially regarding patentable inventions, designs, or trade secrets—staff and students must exercise caution in choosing AI tools. Secure generative AI tools, which operate offline and do not share data externally, should be employed in these cases, avoiding public or unsecured platforms. For guidance on secure generative AI tools, please refer to paragraph 7.9 in the Implementation section below.

6.11. Ongoing education and adaptation

UKZN recognises the rapid evolution of AI technologies and emphasises the importance of ongoing education and adaptation within the academic community. To ensure ethical, effective, and best-practice use of generative AI in academic work, UKZN will (a) revise these *AI Academic Guidelines* on a regular annual basis, and (b) implement initiatives such as establishing a generative AI resource hub and conducting regular workshops. These workshops will offer training at various levels to cater to beginners, intermediate users, and advanced AI users across disciplines.

7. Implementation

7.1. Oversight

Given that these *AI Academic Guidelines* relate to both teaching and learning and research, both the DVC: Teaching and Learning (DVC: T&L) and the DVC: Research and Innovation (DVC: R&I) are the oversight custodians of these *AI Academic Guidelines.* They will oversee the implementation of these *AI Academic Guidelines* at an institutional level, and coordinate the efforts of the College Deans of Teaching and Learning and the College Deans of Research, as provided for in paragraph 7.3 below.

7.2. Administration

The office of the DVC: T&L will be responsible for the administration and oversight of these *AI Academic Guidelines*. This includes, inter alia, that the content of these *AI Academic Guidelines* and any updates or revisions are communicated effectively to the UKZN academic community.

7.3. Implementation custodians

The College Deans of Teaching and Learning and the College Deans of Research are the implementation custodians of these *AI Academic Guidelines*. They will ensure that the *AI Academic Guidelines* are consistently applied across all Schools, provide guidance on the ethical and effective use of generative AI, and in general support their colleagues in implementing generative AI tools in academic work. In addition, they will reach out to their respective industries and/or professions, to ascertain the expectations of these bodies with regard to generative AI literacy and skill of graduates in their fields.

7.4. Thought leadership in generative AI

An AI Thought Leadership Forum is hereby established, consisting of representative from each School, nominated by the Dean and Head of School, and any other staff members invited by the DVC: T&L or DVC: R&I. The AI Thought Leadership Forum will serve as a platform for discussion and thought leadership on the use of generative AI in academic work. In fulfilling this mandate, it may also engage with entities and individuals outside UKZN to inform and enrich its discussions. It will also monitor and consider the latest developments in AI technology, best practices, and ethical considerations including the environmental impact of AI, providing recommendations for potential revisions to these *AI Academic Guidelines*. The AI Thought Leadership Forum will also explore how UKZN can enhance ongoing training, leverage AI research, and position itself as a leader in the responsible use of generative AI on national, continental, and global stages.

7.5. Development of supporting and training materials

The DVC: T&L and DVC: R&I will be responsible for developing comprehensive supporting and training materials within three months of the effective date of these guidelines. The support materials will include a user manual for these *AI Academic Guidelines* that will provide, *inter alia*, standard prompts, as well as standard acknowledgements of the use of generative AI that can be used as is, or adapted by each School. The training material will include workshops, seminars, and video tutorials designed to educate academic staff and students on the ethical and effective use of generative AI in academic work. The training will cater to different levels of generative AI proficiency, ensuring that all academic staff and all students can benefit from these resources.

7.6. Establishment of digital infrastructure

Within three months of the effective date of these guidelines, the office of the DVC: T&L will set up the necessary digital infrastructure to support the implementation of these

guidelines. This infrastructure will include, at a minimum, a dedicated website to provide ongoing support and information on the use of AI in academic work.

7.7. Integration of generative AI skills into the curriculum

7.7.1. School generative AI skills integration plan

Within three months of the effective date of these guidelines, each School must submit a generative AI skills integration plan to the DVC: T&L. This plan must outline:

1. **Approach to integration:** Whether the School will implement self-standing generative AI literacy and skills modules, integrate AI content into existing modules, or adopt a combination of both approaches.
2. **Curriculum content:** The specific generative AI topics and skills that will be covered, including how these will be contextualised within the discipline and aligned with ethical considerations and responsible use.
3. **Staff training:** Plans for training and supporting academic staff in integrating generative AI into their teaching, including proposed workshops, seminars, and other professional development activities.
4. **Implementation timeline:** A detailed timeline for the roll-out of generative AI content within the curriculum, including milestones for developing, testing, and fully integrating the content.
5. **Assessment and evaluation:** The methods and criteria that will be used to assess the effectiveness of the generative AI integration, including how student learning outcomes will be measured and how the integration plan will be adjusted based on feedback and technological advancements.
6. **Interdisciplinary collaboration:** Strategies for fostering interdisciplinary collaboration in generative AI projects and research, including any planned initiatives to encourage cross-disciplinary engagement.
7. **Ongoing review and updates:** A plan for the annual review and update of the generative AI curriculum to ensure it remains current with the latest developments in AI technology and pedagogy.

7.7.2. Review of School generative AI skills integration plan

The College Deans of Teaching and Learning and the College Deans of Research will be tasked with reviewing the generative AI skills integration plans developed by the Schools within their respective Colleges, ensuring that they align with the overall goals of these *AI Academic Guidelines* and support the effective and responsible use of AI. The College Deans of Teaching and Learning and the College Deans of

Research may consult with the AI Thought Leadership Forum to provide constructive feedback on these plans.

7.8. Access to AI tools

At present, many generative AI tools are available free of charge. However, this situation might change. UKZN strives to provide all students and academic staff with reasonable access to relevant, useful generative AI tools for use in their academic work. To accomplish this, UKZN should actively seek to realise such access. This may include, but is not limited to:

- Negotiating institutional licenses for generative AI software.
- Providing generative AI-enabled computer labs on campus.
- Offering subsidised or free access to necessary generative AI tools for students who cannot afford them.

The IT Services department, in consultation with the DVC: T&L, DVC: R&I and the AI Thought Leadership Forum, should develop and maintain a strategy to ensure reasonable access to generative AI tools across the university.

7.9. Institutional guidance of secure generative AI

The IT Services department, in consultation with the DVC: T&L, DVC: R&I and the AI Thought Leadership Forum, should develop and maintain a list of secure generative AI tools, and publish such list on the dedicated website contemplated in paragraph 7.6 above.

Appendix A

AI Declaration [Applicable to all student work]

[Delete non-applicable text:]

a) I did not use generative AI in this report/dissertation/thesis.

b) I made use of generative AI in this report/dissertation/thesis in the following areas: predictive modelling, hypothesis generation, methodology, meta-analysis, qualitative data analysis, simulation and scenario planning, to write initial drafts, to draft summaries, to generate ideas, for visualisation, for reference management, for language editing, others (please specify). In all instances, the content generated by AI was fully interrogated for applicability, accuracy of content, and plagiarism. Citations were added to statements where applicable, to assign credit to authors.

c) Where generative AI was used in the methodology of this report/dissertation/thesis, this is indicated in the relevant methodology section, with full details enclosed.

Signed:
[Insert name and student number]